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VNR12C

Virudhunagar District
Common Half Yearly Examination - December 2024

Standard 12
CHEMISTRY
PART - I

Time: 3.00 Hrs. Marks: 70

I. Choose the correct answer and answer all the questions: **15×1=15**

1) Which one of the following reduction is not thermodynamically feasible?
   a) $Cr_2O_3+2A\ell \rightarrow A\ell_2O_3+2Cr$ b) $A\ell_2O_3+2Cr \rightarrow Cr_2O_3+2A\ell$
   c) $3TiO_2 + 4A\ell \rightarrow 2A\ell_2O_3 + 3Ti$ d) none of these
2) Which of the following is not Sp^2 hybridised?
   a) Fullerene b) Graphite c) Diamond d) Graphene
3) **Assertion :** Ce^{4+} is used as an oxidising agent in volumetric analysis
   Reason : Ce^{4+} has the tendency of attaining +3 oxidation state
   a) Both assertion and reason are true but reason is the correct explanation of assertion
   b) Both assertion and reason are true but reason is not the correct explanation of assertion
   c) Assertion is true but reason is false
   d) Both assertion and reason are false
4) The co-ordination number and oxidation state of the complex $Na_2[Ni(EDTA)]$ are respectively
   a) 2, +2 b) 4, +4 c) 6, +2 d) 6, +4
5) The oxidation state of chlorine in $C\ell_2O_7$ is
   a) +6 b) +7 c) +4 d) +5
6) If the Rate constant of the reaction is $5.8\times10^{-2}S^{-1}$, what is the order of the reaction
   a) first order b) zero order c) second order d) third order
7) $C\ell^-$ is the conjugate base of
   a) $HC\ell O_4$ b) $HC\ell$ c) $C\ell O_4^-$ d) $HC\ell O_3$
8) The vaccant space in bcc lattice unit cell is
   a) 32% b) 33% c) 48% d) 26%
9) Pyrogallol is
   a) 1, 2, 4 trihydroxy benzene b) 1, 3, 5 trihydroxy benzene
   c) 1, 3 dihydroxy benzene d) 1, 2, 3 trihydroxy benzene
10) Which of the following electrolytic solution has the least specific conductance
   a) 1N b) 0.01 N c) 0.1 d) 2N
11) Fog is a colloidal solution of
   a) gas in gas b) solid in gas c) gas in liquid d) liquid in gas
12) Which one of the following reduces Tollens reagent
   a) Formic acid b) acetic acid c) Benzophenone d) none of these
13) $C_6H_5NO_2 \xrightarrow{Fe/HC\ell}$, A $\xrightarrow[273K]{NaNO_2/HC\ell}$ B $\xrightarrow{H_2O/283K}$ C, C is
   a) C_6H_5OH b) $C_6H_5CH_2OH$ c) C_6H_5CHO d) $C_6H_5NH_2$
14) Glucose and Mannose are epimers at
   a) C_3 Carbon b) C_4 carbon c) C_1 carbon d) C_2 carbon
15) Match the following:
   1) Major tranquilizer - i) non-steroid anti inflammatory drug
   2) analgesics - ii) Propofol
   3) NSAIDS - iii) Clozapine
   4) Intravenous general anasthetics - iv) Aspirin
   a) 1-iii 2-iv 3-i 4-ii b) 1-i 2-ii 3-iii 4-iv
   c) 1-ii 2-i 3-iv 4-iii d) 1-iv 2-iii 3-ii 4-i

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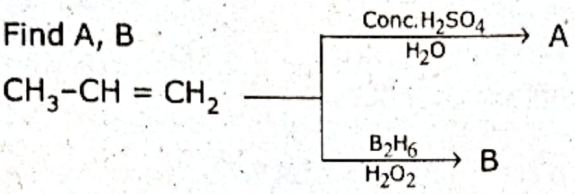
PART - II

II. Answer any six questions. Q.No. 24 is compulsory: **6×2=12**

16) How the ores are concentrated by Gravity separation method?
17) Write the chromyl chloride test.
18) Uses of borax-2
19) Define unit cell
20) Define Faraday's second law of electrolysis.
21) Distinguish antagonists from agonists.
22) Define order of the reaction.
23) How will you convert the following:
    i) Ethylene glycol → 1, 4 dioxane
    ii) Ethylene → Ethylene glycol
24) Find A, B



Part - III

III. Answer any six questions. Q.No. 33 is compulsory: **6×3=18**

25) What is Catenation? Give any two conditions for catenation.
26) Give the laboratory preparation of chlorine.
27) Differences between octahedral voids and tetrahedral voids.
28) Write the Thorpe-Nitrile condensation reaction.
29) What is formalin-Give uses.
30) How will you prepare Nylon-2-nylon-6
31) Write the formula for the following co-ordination compounds.
    a] Sodium tetra fluoridodihydroxydochromate (III)
    b] Pentaamminenitrito-KN cobalt (III) ion
32) Explain the formation of peptide bond with example.
33) A copper electrode is dipped in 0.1 M copper sulphate solution at 25°C. Calculate the electrode potential of copper. [Given $E^{\circ}_{Cu^{2+}/Cu} = 0.34V$]

Part - IV

IV. Answer all the questions: **5×5=25**

34) a] i) Describe the Mond Process. ii) What is roasting?

(OR)

b] Describe the rate constant for the frist order reaction.

35) a] i) Distinguish diamond from graphite.
    ii) McAfee process.

(OR)

b] Write the differences between Lanthanoids and Actinoids.

36) a] Write notes on i) Holmes signal ii) Ziegler-Natta catalyst.

(OR)

b] Write the postulates of Valence Bond Theory.

37) a] i) What are Lewis acids and Lewis bases? Give examples.
    ii) Differences between DNA and RNA - 6 points.

(OR)

b] What is packing efficiency? Calculate the packing efficiency of FCC lattice.

38) a] i) Write briefly Vulcanization of Rubber
    ii) What are promotors. Give examples.

(OR)

b] Explain the mechanism of Cannizaro reaction.

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